

# Probability and Statistics

Nimisha Roy

Lecturer, SCI, College of Computing, Georgia Tech

# Outline

- Probability Distributions 
- Joint and Conditional Probability Distributions
- Bayes' Rule
- Mean and Variance
- Properties of Gaussian Distribution
- Maximum Likelihood Estimation

# Probability

probability of an event =  $\frac{1}{6}$   
unbiased die

- A **sample space S** is the set of all possible outcomes of a conceptual or physical, repeatable experiment. (S can be finite or infinite.)
  - E.g., S may be the set of all possible outcomes of a dice roll: S  
(1 2 3 4 5 6)
  - E.g., S may be the set of all possible nucleotides of a DNA site: S  
(A C G T)
- E.g., S may be the set of all possible time-space positions of an aircraft on a radar screen.
- An **Event A** is any subset of S
  - Seeing "1" or "6" in a dice roll; observing a "G" at a site; UA007 in space-time interval



# Three Key Ingredients in Probability Theory

A **sample space** is a collection of all possible **outcomes**

Random variables  $X$  represents **outcomes** in sample space

Probability of a random variable to happen

$$p(x) = p(X = x)$$

probability  
that r.v.  
 $p(x) \neq 0$   
outcome  $x$

random  
variable

a particular  
outcome

## Continuous variable

pdf & pmf are probability distribution function = PDF

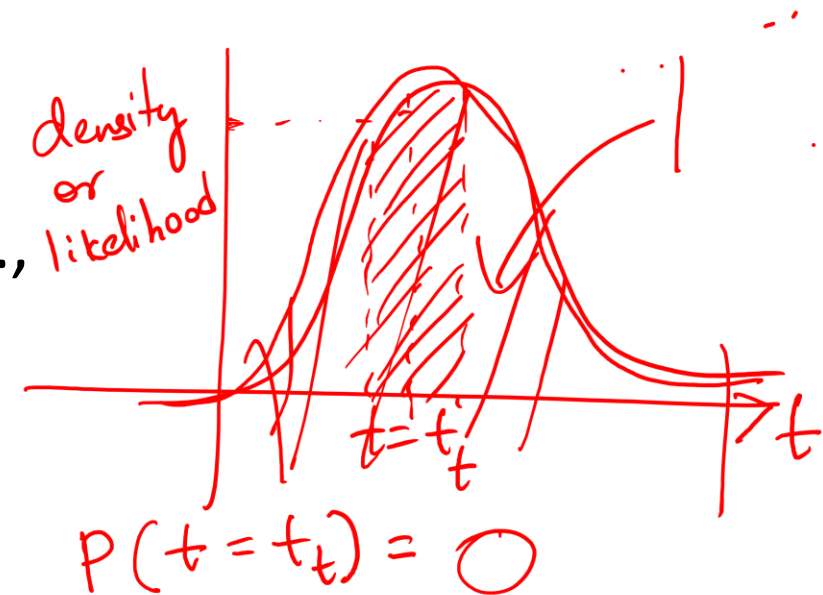
**Definition:** Takes values from a continuous range (e.g., any real number within an interval).

**Distribution:** Governed by a probability density function (PDF).

### Key Concepts:

- The **density** represents likelihood, but not actual probability at a specific point.
- Example: **Temperature**, which can be any real value (e.g., 72.3°F).
- Common distribution: Gaussian (Normal) Distribution.

$$\int_x p(x) dx = 1$$



## Discrete variable

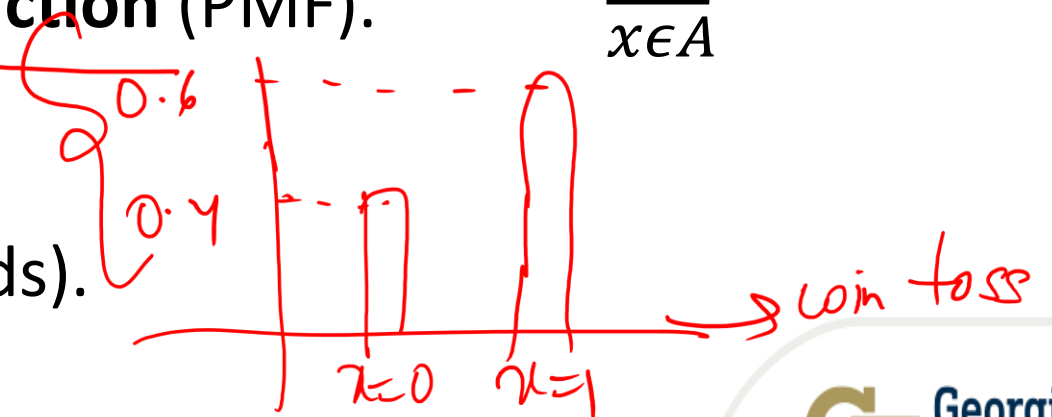
**Definition:** Takes values from a countable set (e.g., integers).

**Distribution:** Governed by a probability mass function (PMF).

### Key Concepts:

- The function directly gives **probability values**.
- Example: **A coin flip** (e.g., 0 for tails, 1 for heads).
- Common distribution: **Bernoulli Distribution**.

$$\sum_{x \in A} p(x) = 1$$



# Continuous Probability Functions

is to optimize objective function  $\rightarrow$  parameters

training  $\approx$  optimization

## Examples:

### Uniform Density Function: $(a, b \in \Theta)$

$$f_x(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

*objective function*  $\rightarrow$

$\frac{1}{c-d}$  for  $a \leq x \leq b$   
 $c=b$   
 $d=a$



### Exponential Density Function:

$$f_x(x) = \frac{1}{\mu} e^{-\frac{x}{\mu}} \quad \text{for } x \geq 0$$

$\mu \in \Theta$

In our class,  $\mu$  means  
 expected value or  
 weighted average  
 AND arithmetic mean

### Gaussian(Normal) Density Function

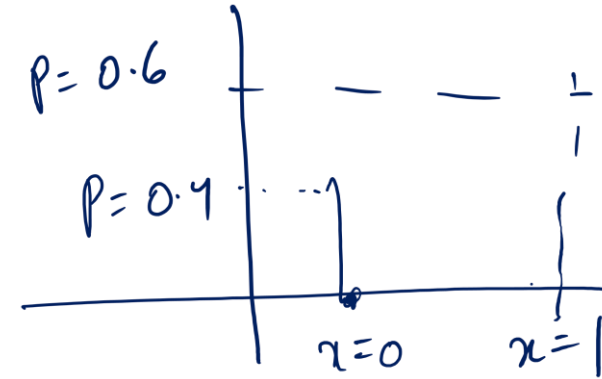
$$f_x(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$\mu, \sigma \in \Theta$

$\frac{1}{\sqrt{2\pi}b}$

$e^{-\frac{(x-a)^2}{2b}}$

# Discrete Probability Functions



- Examples:

- Bernoulli Distribution:

- $$\begin{cases} 1 - p & \text{for } x = 0 \\ p & \text{for } x = 1 \end{cases}$$

In Bernoulli, just a **single** trial is conducted

- Binomial Distribution:

- $$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

**k** is number of successes

**n-k** is number of failures

$\binom{n}{k}$  The total number of ways of selection **k** distinct combinations of **n** trials, **irrespective of order**.

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- Joint and Conditional Probability Distributions
- Bayes' Rule
- Mean and Variance
- Properties of Gaussian Distribution
- Maximum Likelihood Estimation



# Example



X = Throw a  
dice



Y = Flip a coin

**X** and **Y** are random variables

**N** = total number of trials



$n_{ij}$  = Number of occurrence

**Y**  
 $y_{j=2} = tail$   
 $y_{j=1} = head$

|              | $x_{i=1} = 1$ | $x_{i=2} = 2$ | $x_{i=3} = 3$ | $x_{i=4} = 4$ | $x_{i=5} = 5$ | $x_{i=6} = 6$ | $C_j$ |
|--------------|---------------|---------------|---------------|---------------|---------------|---------------|-------|
| $n_{ij} = 3$ | $n_{ij} = 4$  | $n_{ij} = 2$  | $n_{ij} = 5$  | $n_{ij} = 1$  | $n_{ij} = 5$  | 20            |       |
| $n_{ij} = 2$ | $n_{ij} = 2$  | $n_{ij} = 4$  | $n_{ij} = 2$  | $n_{ij} = 4$  | $n_{ij} = 1$  | 15            |       |
| $C_i$        | 5             | 6             | 6             | 7             | 5             | 6             | N=35  |

**X** $C_j$ 
 $x_{i=1} = 1$     $x_{i=2} = 2$     $x_{i=3} = 3$     $x_{i=4} = 4$     $x_{i=5} = 5$     $x_{i=6} = 6$ 
**Y** $y_{j=2} = \text{tail}$  $y_{j=1} = \text{head}$  $C_i$ 

|              |              |              |              |              |              |      |
|--------------|--------------|--------------|--------------|--------------|--------------|------|
| $n_{ij} = 3$ | $n_{ij} = 4$ | $n_{ij} = 2$ | $n_{ij} = 5$ | $n_{ij} = 1$ | $n_{ij} = 5$ | 20   |
| $n_{ij} = 2$ | $n_{ij} = 2$ | $n_{ij} = 4$ | $n_{ij} = 2$ | $n_{ij} = 4$ | $n_{ij} = 1$ | 15   |
| 5            | 6            | 6            | 7            | 5            | 6            | N=35 |

joint prob.  
 $p(x=1, y=t) = \frac{3}{35} = \frac{n_{ij}}{N}$

marginal prob.  
 $p(x=4) =$

$$\frac{(5+2)}{35} = \frac{C_i}{N}$$

SUM RULE

$$p(x) = \sum_y P(x, y)$$

$$p(y=t) = \frac{20}{35} = \frac{C_j}{N}$$

$$p(x=3 | y=t) = \frac{2}{20} = \frac{n_{ij}}{C_j}$$

$$p(y=t | x=3) = \frac{2}{6} = \frac{n_{ij}}{C_i}$$

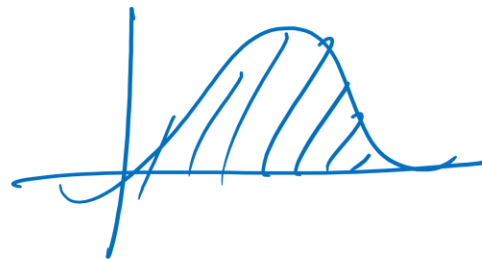
# Recap

- Sample Space and Event

set of all possible outcomes

1 particular outcome

- Probability Distribution Function



$p(x)$  or  $p(x=a)$   
 random variable →  $x$   
 event →  $a$

continuous → probability density function  
 discrete → probability mass function

- Objective Function

$$f(x)$$

has parameters  $\theta$

Gaussian dist.  $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$   
 $\{\mu, \sigma\} \in \theta$

- Joint Probability, Marginal Probability & Conditional Probability

$$p(x, y)$$

$$p(a, b, c, \dots, n)$$

$$p(x)$$

$$p(x|y)$$

shrinking sample space

# Example



X = Throw a  
dice



Y = Flip a coin

**X** and **Y** are random variables

**N** = total number of trials

$n_{ij}$  = Number of occurrence

|          |                  | <b>X</b>      |               |               |               |               |               |       |
|----------|------------------|---------------|---------------|---------------|---------------|---------------|---------------|-------|
|          |                  | $x_{i=1} = 1$ | $x_{i=2} = 2$ | $x_{i=3} = 3$ | $x_{i=4} = 4$ | $x_{i=5} = 5$ | $x_{i=6} = 6$ | $C_j$ |
| <b>Y</b> | $y_{j=2} = tail$ | $n_{ij} = 3$  | $n_{ij} = 4$  | $n_{ij} = 2$  | $n_{ij} = 5$  | $n_{ij} = 1$  | $n_{ij} = 5$  | 20    |
|          | $y_{j=1} = head$ | $n_{ij} = 2$  | $n_{ij} = 2$  | $n_{ij} = 4$  | $n_{ij} = 2$  | $n_{ij} = 4$  | $n_{ij} = 1$  | 15    |
| $C_i$    |                  | 5             | 6             | 6             | 7             | 5             | 6             | N=35  |

**Y** $y_{j=2} = \text{tail}$  $y_{j=1} = \text{head}$  $C_i$ 

|  | $x_{i=1} = 1$ | $x_{i=2} = 2$ | $x_{i=3} = 3$ | $x_{i=4} = 4$ | $x_{i=5} = 5$ | $x_{i=6} = 6$ | $C_j$       |
|--|---------------|---------------|---------------|---------------|---------------|---------------|-------------|
|  | $n_{ij} = 3$  | $n_{ij} = 4$  | $n_{ij} = 2$  | $n_{ij} = 5$  | $n_{ij} = 1$  | $n_{ij} = 5$  | 20          |
|  | $n_{ij} = 2$  | $n_{ij} = 2$  | $n_{ij} = 4$  | $n_{ij} = 2$  | $n_{ij} = 4$  | $n_{ij} = 1$  | 15          |
|  | 5             | 6             | 6             | 7             | 5             | 6             | <u>N=35</u> |

**X**

$$p(y=t, x=4) = 5/35 = n_{ij}/N$$

$$p(y=t) = 20/35 = C_j/N$$

summing over all X

$$p(y=t | x=4) = \frac{5}{7} = \frac{n_{ij}}{C_i}$$

$$p(x=4 | y=t) = \frac{5}{20} = \frac{n_{ij}}{C_j}$$

$$p(x|y) = \frac{n_{ij}}{C_j} = \frac{n_{ij}}{N} \cdot \frac{N}{C_j} = \frac{p(x,y)}{p(y)}$$

$$\begin{aligned} p(x,y) &= p(y|x)p(x) \\ &= p(x|y)p(y) \end{aligned}$$

$$= \sum_x p(x,y) = p(y)$$

SUM RULE

$x$  is a more informative r.v.

$$\Rightarrow \underline{p(x|y) p(y) = p(x,y)}$$

PRODUCT RULE

## Probability:

$$p(X = x_i) = \frac{c_i}{N}$$

## Joint probability:

$$p(X = x_i, Y = y_j) = \frac{n_{ij}}{N}$$

## Conditional probability:

$$p(Y = y_j | X = x_i) = \frac{n_{ij}}{c_i}$$

Do not like list

① for loops

② inverse

③ matrix multiplication

④ joint probability

## Sum rule

$$p(X = x_i) = \sum_{j=1}^L p(X = x_i, Y = y_j) \Rightarrow p(X) = \sum_Y P(X, Y)$$

## Product rule

$$p(X = x_i, Y = y_j) = \frac{n_{ij}}{N} = \frac{n_{ij}}{c_i} \frac{c_i}{N} = p(Y = y_j | X = x_i) p(X = x_i)$$

$$p(X, Y) = p(Y|X)p(X)$$

# Conditional Independence

- Examples:

$$P(\text{Virus} | \text{DrinkBeer}) = P(\text{Virus})$$

iff Virus is independent of Drink Beer

$$P(\text{Flu} | \text{Virus}, \text{DrinkBeer}) = P(\text{Flu} | \text{Virus})$$

iff Flu is independent of Drink Beer, given Virus

$$P(\text{Headache} | \text{Flu}, \text{Virus}, \text{DrinkBeer}) = P(\text{Headache} | \text{Flu}, \text{DrinkBeer})$$

iff Headache is independent of Virus, given Flu and Drink Beer

$$\begin{aligned} P(h, f, v, d) &= P(h | f, v, d) P(f, v, d) \\ &= P(h | f, d) P(f | v, d) P(v, d) \end{aligned}$$

$$= P(h | f, d) P(f | v) P(v | d) P(d)$$

$$= P(h | f, d) P(f | v) P(v) P(d)$$

Assume the above independence, we obtain:

$$\begin{aligned} &P(\text{Headache}, \text{Flu}, \text{Virus}, \text{DrinkBeer}) \\ &= P(\text{Headache} | \text{Flu}, \text{Virus}, \text{DrinkBeer}) P(\text{Flu} | \text{Virus}, \text{DrinkBeer}) \\ &\quad P(\text{Virus} | \text{DrinkBeer}) P(\text{DrinkBeer}) \\ &= P(\text{Headache} | \text{Flu}, \text{DrinkBeer}) P(\text{Flu} | \text{Virus}) P(\text{Virus}) P(\text{DrinkBeer}) \end{aligned}$$

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- Bayes' Rule 
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# Bayes' Rule

product rule

$$P(x, y) = P(x|y) \cdot P(y)$$

- $P(X|Y)$  = Fraction of the worlds in which  $X$  is true given that  $Y$  is also true.

$$\underline{P(x|y)} = \frac{P(x, y)}{P(y)}$$

- For example:

- $H$  = "Having a headache"
- $F$  = "Coming down with flu"
- $P(\text{Headache}|\text{Flu})$  = fraction of flu-inflicted worlds in which you have a headache. How to calculate?

$$P(x|y) = \frac{P(y|x)P(x)}{P(y)}$$

- Definition:

$$P(X|Y) = \frac{P(X, Y)}{P(Y)} = \frac{P(Y|X)P(X)}{P(Y)}$$

Corollary:

$$P(X, Y) = P(Y|X)P(X)$$

This is called **Bayes Rule**

# Bayes' Rule

- $$P(\text{Headache}|\text{Flu}) = \frac{P(\text{Headache}, \text{Flu})}{P(\text{Flu})}$$

$$= \frac{P(\text{Flu}|\text{Headache})P(\text{Headache})}{P(\text{Flu})}$$

Other cases:

- $$P(Y|X) = \frac{P(X|Y)P(Y)}{P(X|Y)P(Y) + P(X|\neg Y)P(\neg Y)}$$

- $$P(Y = y_i|X) = \frac{P(X|Y)P(Y)}{\sum_{i \in S} P(X|Y = y_i)P(Y = y_i)}$$

- $$P(Y|X, Z) = \frac{P(X|Y, Z)P(Y, Z)}{P(X|Y, Z)P(Y, Z) + P(X|\neg Y, Z)P(\neg Y, Z)}$$

when  $y$  is a binary variable

$$\begin{aligned}
 P(y|x) &= \frac{P(x|y)P(y)}{P(x)} \\
 &= \frac{P(x|y)P(y)}{\sum_y P(x, y)} \\
 &= \frac{P(x|y)P(y)}{P(x, y) + P(x, \neg y)} \\
 &= \frac{P(x|y)P(y)}{P(x|y)P(y) + P(x|\neg y)P(\neg y)}
 \end{aligned}$$

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In ML  $\rightarrow$  Expectation or weighted average is the same as arithmetic mean

# Mean and Variance

- Expectation: The mean value, center of mass, first moment:

$$E_X[g(X)] = \int_{-\infty}^{\infty} \underbrace{g(x)} \underbrace{p_X(x)} dx = \mu$$

$E[g(x)]$

discrete

$$E[g(x)] = \sum_{i=1}^n g(x_i) p(x_i)$$

- N-th moment:  $g(x) = x^n$

- N-th central moment:  $g(x) = (x - \mu)^n$

- Mean:  $E_X[X] = \int_{-\infty}^{\infty} x p_X(x) dx$

- $E[\alpha X] = \alpha E[X]$

- $E[\alpha + X] = \alpha + E[X]$

- Variance(Second central moment):  $Var(x) =$

$$E_X[(X - E_X[X])^2] = E_X[X^2] - E_X[X]^2$$

- $Var(\alpha X) = \alpha^2 Var(X)$

- $Var(\alpha + X) = Var(X)$

$$\underline{\underline{E[(x - E[x])^2]}}$$

# Expected value and average

biased 3 sided die

$$x = [1, 2, 3]$$

limited data

$$p(x) = \left[ \frac{1}{6}, \frac{1}{3}, \frac{1}{2} \right]$$

add up to 1.  
valid pmf ??

$$g(x) = x$$

$$E[g(x)] = \sum_{i=1}^3 g(x_i) p(x_i)$$

$$= 1 \times \frac{1}{6} + 2 \times \frac{1}{3} + 3 \times \frac{1}{2} = \frac{14}{6}$$

$$\mu = \frac{1+2+3}{3} = 2$$

We roll this biased die 6 times

$$x = [1, 2, 2, 3, 3, 3]$$

$$\mu = \frac{1+2+2+3+3+3}{6} = \frac{14}{6}$$

Expectation determined by probability distribution. Arithmetic average determined by observed outcomes of trials.

Both are the same in our class because we assume we have sufficient data that models the distribution. So, the arithmetic average is a good estimate of the true expectation (Law of Large Numbers).

# Variance and average:

how much our data varies from the mean

$$X = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{matrix} \text{height} \\ n \times d \\ 3 \end{matrix}$$

$$\text{Variance} = \frac{1}{N} \sum_{i=1}^n (x - \mu)^2$$

for loop

$$= E[x - \mu]^2 = E[x - E[x]]^2$$

mean

$$\text{Var} = \frac{1}{n} \bar{X}^T \bar{X}$$

scalar

$\bar{X}$  is the centered matrix

$$\bar{X}_{3 \times 1}$$

$$\bar{X} = \begin{bmatrix} 1 - \mu_h \\ 2 - \mu_h \\ 3 - \mu_h \end{bmatrix} \frac{1}{N} \bar{X}^T \bar{X}$$

$$\text{Var} = \frac{1}{N} \begin{bmatrix} 1 - \mu_h & 2 - \mu_h & 3 - \mu_h \end{bmatrix} \begin{bmatrix} 1 - \mu_h \\ 2 - \mu_h \\ 3 - \mu_h \end{bmatrix}$$

$$= \frac{1}{N} \left( (1 - \mu_h)^2 + (2 - \mu_h)^2 + (3 - \mu_h)^2 \right)$$

1x1 scalar because we have only 1 feature.

If d features,  $\text{Var}_{d \times d} \rightarrow \text{Covariance matrix}$

# Covariance:

If we have  $d$  features, we can compute covariance matrix of size  $d \times d$ .

$$\text{Cov}(\Sigma) = \frac{1}{N} \bar{X}^T \bar{X}$$

$$X = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}_{n \times d}$$

3      2

$$\bar{X} = \begin{bmatrix} 1 - \mu_h & 4 - \mu_w \\ 2 - \mu_h & 5 - \mu_w \\ 3 - \mu_h & 6 - \mu_w \end{bmatrix}$$

$$\frac{1}{N} \begin{bmatrix} 1 - \mu_h & 2 - \mu_h & 3 - \mu_h \\ 4 - \mu_w & 5 - \mu_w & 6 - \mu_w \end{bmatrix} \begin{bmatrix} 1 - \mu_h & 4 - \mu_w \\ 2 - \mu_h & 5 - \mu_w \\ 3 - \mu_h & 6 - \mu_w \end{bmatrix}$$

1st term:  $(1 - \mu_h)^2 + (2 - \mu_h)^2 + (3 - \mu_h)^2$   
 4th term:  $(4 - \mu_w)^2 + (5 - \mu_w)^2 + (6 - \mu_w)^2$

Cov =

h  
w

high no.      high no.

$$\sigma_{wh} = (1 - \mu_h)(4 - \mu_w) + \dots$$

$\sigma_{wh}$  tells us how weight & height correlate

$\sigma_{wh} = \sigma_{hw}$  ? YES      Cov  $\rightarrow$  Symmetrical

$\sigma_{wh} = 0$        $\sigma_{wh} > 0$

# Correlation:

$$\text{Cov} = \frac{1}{N} \bar{X}^T \bar{X}$$

$$\text{Cor} = \frac{1}{N} \bar{X}^{*T} \bar{X}^*$$

$$\bar{X}^* = \frac{\bar{X}}{\sigma}$$

$\sigma^2 \leftarrow$  variance  
 $\sigma \rightarrow$  std. dev

Normalize our features  
 $\rightarrow [0, 1] \checkmark \times$   
 $[-1, 1]$   
 standardization

$$\bar{X}^* = \begin{bmatrix} \frac{1-\mu_h}{\sigma_h} & \frac{4-\mu_w}{\sigma_w} \\ \frac{2-\mu_h}{\sigma_h} & \frac{5-\mu_w}{\sigma_w} \\ \frac{3-\mu_h}{\sigma_h} & \frac{6-\mu_w}{\sigma_w} \end{bmatrix}$$

$$\text{Cor}[1,1] = \frac{1}{N} \begin{bmatrix} \frac{1-\mu_h}{\sigma_h} & \frac{2-\mu_h}{\sigma_h} & \frac{3-\mu_h}{\sigma_h} \\ \frac{4-\mu_w}{\sigma_w} & \frac{5-\mu_w}{\sigma_w} & \frac{6-\mu_w}{\sigma_w} \end{bmatrix} \begin{bmatrix} \frac{1-\mu_h}{\sigma_h} & 4-\mu_w \\ \frac{2-\mu_h}{\sigma_h} & \\ \frac{3-\mu_h}{\sigma_h} & \end{bmatrix}$$

$$\frac{1}{N} \left( \frac{(1-\mu_h)^2}{\sigma_h^2} + \frac{(2-\mu_h)^2}{\sigma_h^2} + \frac{(3-\mu_h)^2}{\sigma_h^2} \right) = \frac{\sigma_h^2}{\sigma_h^2} = 1$$

$$\text{Cor} = \begin{bmatrix} 1 & \dots \\ \dots & 1 \end{bmatrix} \begin{matrix} h \\ w \end{matrix}$$

Useful in EDA. If features are correlated, then data may have redundancy.

# Uncorrelated vs Independent RV

**Uncorrelated** (Definition:  $\text{Cov}(X,Y)=0$ .)

- Means no **linear relationship** between  $X$  and  $Y$ .
- Does **not** rule out non-linear dependence.
- Example:  $X \sim U(-1,1)$ ,  $Y=x^2 \rightarrow$  Uncorrelated but dependent.

**Independent** (Definition:  $P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$ )

- Stronger condition: knowing one variable gives **no information** about the other.
- Independence  $\Rightarrow$  Uncorrelated (if variances exist).

## Key Differences

- Independence is a **stronger** property.
- Uncorrelated only removes **linear relationships**.
- Uncorrelated  $\nRightarrow$  Independent.

## ML Relevance

- Most ML models care about uncorrelatedness because they only model linear relationships.
- True independence is rarer and much harder to check, but it's what we assume in stronger probabilistic models

# For Joint Distributions

- Expectation and Covariance:

- $E[X + Y] = E[X] + E[Y]$

- $\text{cov}(X, Y) = E[(X - E_X[X])(Y - E_Y(Y))] = E[XY] - E[X]E[Y]$

- $\text{Var}(X + Y) = \text{Var}(X) + 2\text{cov}(X, Y) + \text{Var}(Y)$

# Recap

- Sum and Product Rule. Bayes Rule

$$p(a) = \sum_b p(a, b)$$

$$p(a, b) = p(a|b) p(b) = p(b|a) p(a)$$

$$p(a|b) = \frac{p(b|a) p(a)}{p(b)}$$

- Expectation and arithmetic mean

$$E[g(x)] = \sum_{i=1}^n g(x_i) p(x_i) \quad \left. \vphantom{\sum_{i=1}^n} \right\} = \text{Average or arithmetic mean}$$

- Variance

$$\frac{1}{N} \sum_{i=1}^n (x_i - \mu)^2 \quad \left. \vphantom{\sum_{i=1}^n} \right\} \text{ only if we have } \underline{1} \text{ features}$$

- Covariance

if we have  $d$  vectors or features

$$\begin{bmatrix} \sigma_h^2 & \sigma_{hw} \\ \sigma_{wh} & \sigma_w^2 \end{bmatrix}_{d \times d}$$

$$= \frac{1}{N} \bar{X}^T \bar{X} \quad \leftarrow \text{centered}$$

$\sigma_{hw} = \sigma_h^2$  w & h are same

- Correlation

scale  $\rightarrow$  standardization

$$\bar{X}^* = \bar{X} / \sigma$$

$$\begin{bmatrix} 1 & 1 \\ \vdots & \vdots \end{bmatrix}_{d \times d}$$

$$\text{Cor} = \frac{1}{N} \bar{X}^{*T} \bar{X}^*$$

$$-1 < < 1$$

- Uncorrelated vs Independent r.v.

could be non-linearly dependence

$$p(x, y) = p(x) p(y)$$

$$p(x|y) = p(x)$$

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# Gaussian Distribution

- Gaussian Distribution:

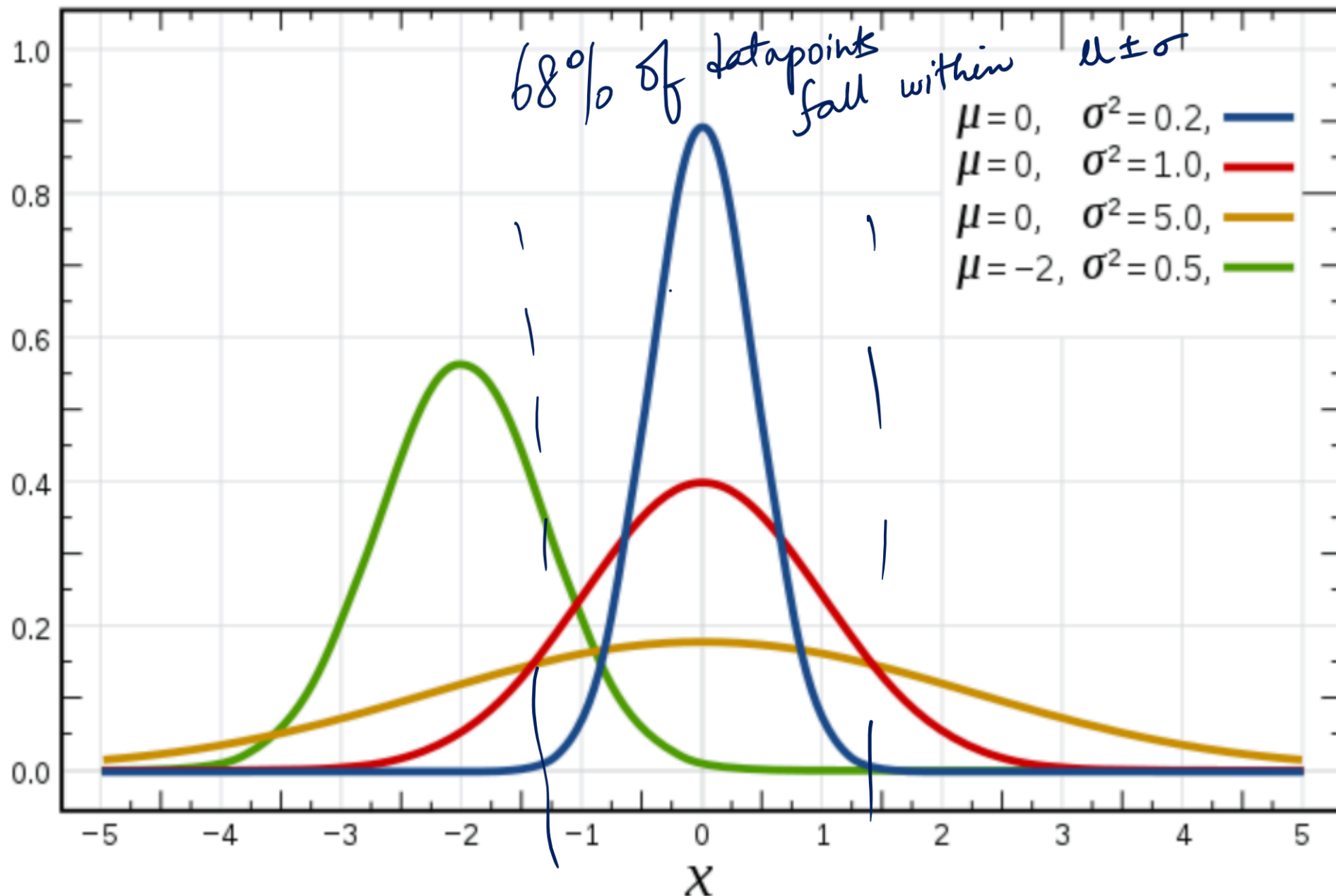
$f(x)$   $\uparrow$  optimization function  
what  $\theta$  is?

$$\mathcal{Z} = (\mu, \sigma^2)$$

$$f(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$a = \mu$   
 $b = \sigma^2$

Probability density function



Probability versus likelihood

# Multivariate Gaussian Distribution

$$p(x|\mu, \Sigma) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp\left\{-\frac{1}{2} (x - \mu)^\top \Sigma^{-1} (x - \mu)\right\}$$

- Moment Parameterization  $\mu = E(X)$

$$\Sigma = \text{Cov}(X) = E[(X - \mu)(X - \mu)^\top]$$

- Mahalanobis Distance  $\Delta^2 = (x - \mu)^\top \Sigma^{-1} (x - \mu)$
- Tons of applications (MoG, FA, PPCA, Kalman filter,...)

# Properties of Gaussian Distribution

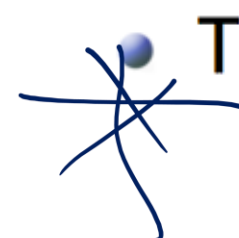
- The **linear transform** of a Gaussian r.v. is a Gaussian. Remember that no matter how  $x$  is distributed

$$\overset{\text{Gaussian r.v.}}{E(\underline{AX} + \underline{b})} = AE(X) + b$$

$$\text{Cov}(AX + b) = A\text{Cov}(X)A^T$$

this means that for Gaussian distributed quantities:

$$\underline{X \sim N(\mu, \Sigma)} \rightarrow \underline{AX + b \sim N(A\mu + b, A\Sigma A^T)}$$

-  The **sum** of two independent Gaussian r.v. is a Gaussian  
 $Y = X_1 + X_2, X_1 \perp X_2 \rightarrow \mu_y = \mu_1 + \mu_2, \Sigma_y = \Sigma_1 + \Sigma_2$

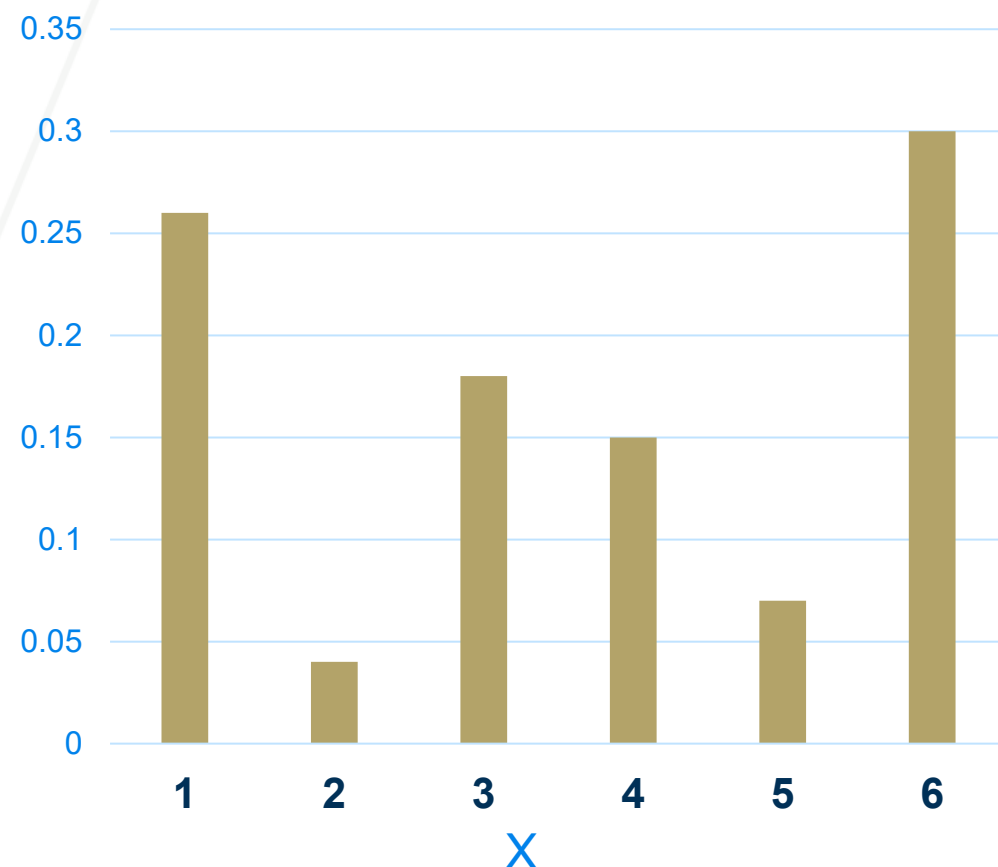
- The **multiplication** of two Gaussian functions is another Gaussian function (although no longer normalized)

$$N(a, A)N(b, B) \propto N(c, C),$$

$$\text{where } C = (A^{-1} + B^{-1})^{-1}, c = CA^{-1}a + CB^{-1}b$$

# Central Limit Theorem

Probability mass function of a **biased** dice



Let's say, I am going to get a sample from this pmf having a size of  **$n = 4$**

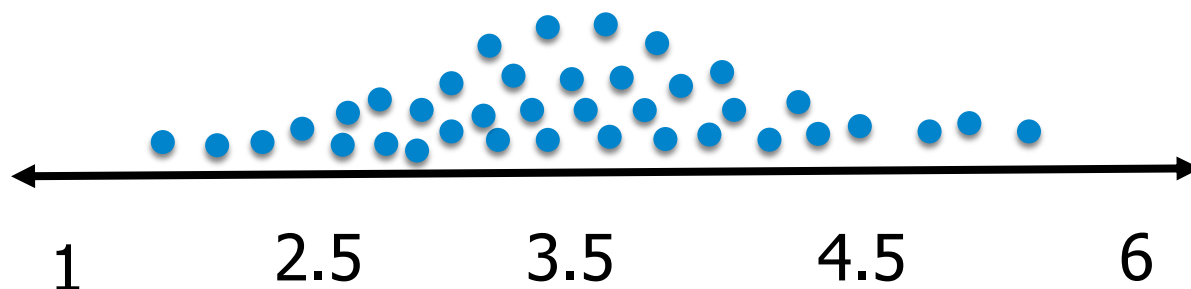
$$S_1 = \{1,1,1,6\} \Rightarrow E(S_1) = 2.25$$

$$S_2 = \{1,1,3,6\} \Rightarrow E(S_2) = 2.75$$

$\vdots$

$$S_m = \{1,4,6,6\} \Rightarrow E(S_m) = 4.25$$

- According to CLT, if you sample enough from any distribution with finite variance you will get an approximate Gaussian distribution.
- No matter what the population looks like, the average of many samples looks Normal.
- Explains **why the Normal distribution is everywhere** in statistics and ML.



# Prob vs Likelihood

*Probability predicts data from parameters.*

*Likelihood evaluates parameters from data.*

## •Probability

- Forward direction: given parameters, what's the chance of data?
- $P(data | \theta)$
- Example: *If the coin has bias  $\theta = 0.7$ , what's the probability of 8 heads in 10 tosses?*
- Varies with **different outcomes** of data.

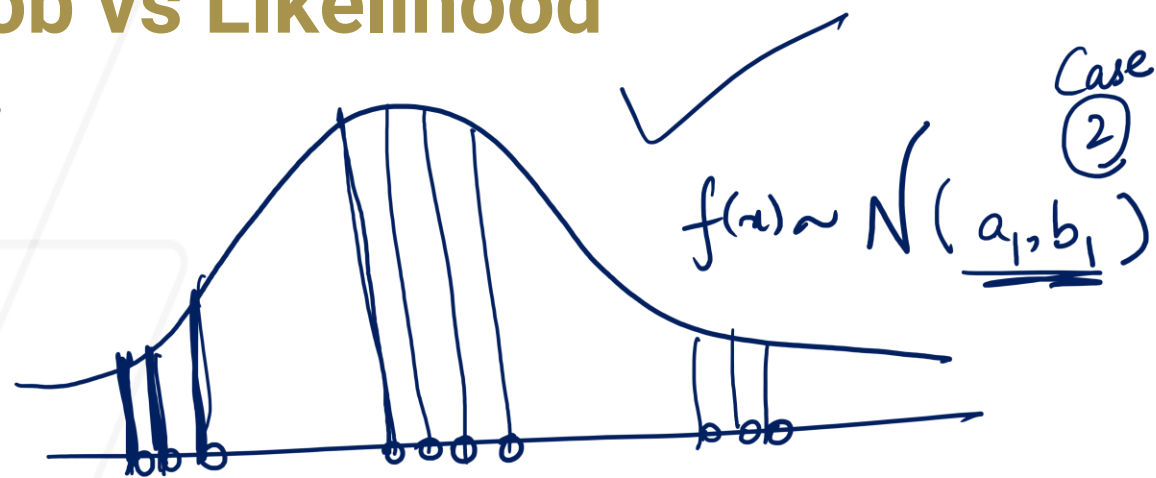
Binomial  $\frac{{}^{10}C_8 (0.7)^8 (0.3)^2}$

## •Likelihood

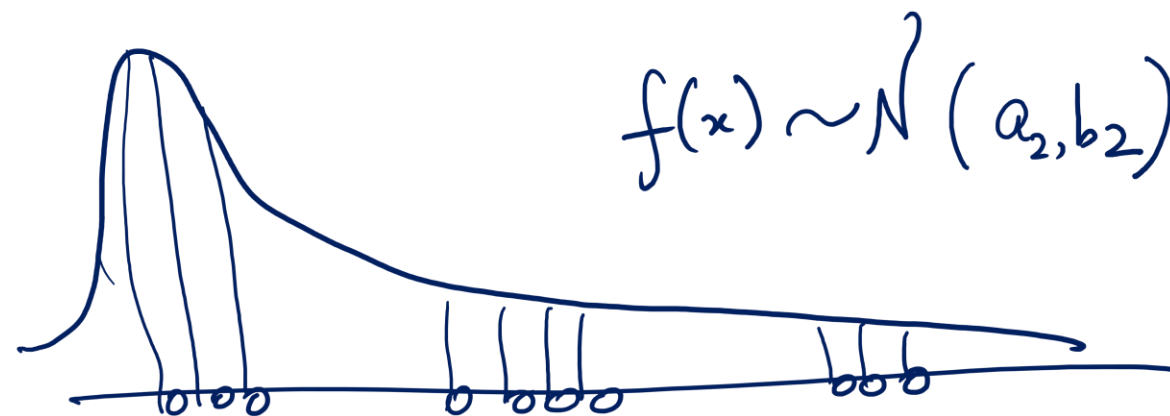
- Reverse view: given data, how plausible are parameters?
- $L(\theta | data) \propto P(data | \theta)$
- Example: *I observed 8 heads in 10 tosses. How likely is it that the coin's bias is  $\theta = 0.7$ ?*
- Varies with **different parameter values**.

# Prob vs Likelihood

Case ①



Case ②



Likelihood tells us which is the best fit of prob. dist. function that describes our observed data.

$$\theta = \{a, b\}$$

$$= f(x|\theta) = \frac{p(x|\theta)}{n=10}$$

Likelihood

$$L(\theta|x)$$

parameters

datapoints

$$L(\underline{a, b}|x) = p(x^{\{1\}}, x^{\{2\}}, x^{\{3\}}, \dots, x^{\{n\}} | a, b)$$

datapoint

i.i.d.  
independently & identically distributed

$$L(a, b|x) =$$

$$\prod_{i=1}^n$$

$$p(x^{\{i\}} | a, b) \dots$$

# Prob vs Likelihood

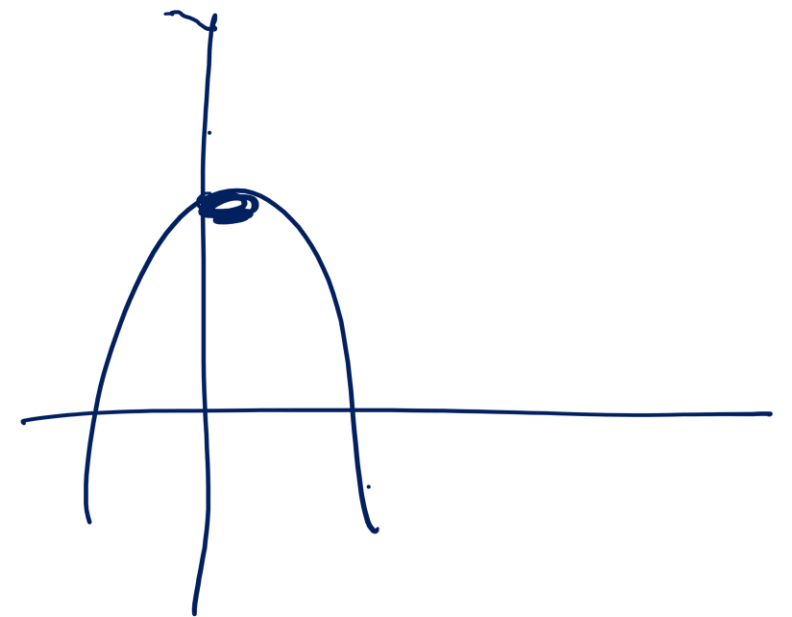
$$\log(a \cdot b \cdot c) = \log a + \log b + \log c$$

$$L(a, b | x) = \prod_{i=1}^n p(x^{(i)} | a, b)$$

$$\log L(a, b | x) = \sum_{i=1}^n \log p(x^{(i)} | a, b)$$

$\log L(a, b | x)$   
↓  
 $l$  (log likelihood)

$$\max_{a, b} l(a, b | x) = \sum_{i=1}^n \log p(x^{(i)} | a, b)$$



$$\frac{\partial l}{\partial a} = 0$$

$$\frac{\partial l}{\partial b} = 0$$

$\{a, b\}$  optimal values

$a = \mu$   
 $b = \sigma^2$

max log likelihood  
min - log likelihood

MLE: Max.  
Likelihood  
Estimation

# Outline

- Probability Distributions
- Joint and Conditional Probability Distributions
- Bayes' Rule
- Mean and Variance
- Properties of Gaussian Distribution
- Maximum Likelihood Estimation 

# Maximum Likelihood Estimation

- Probability: inferring probabilistic quantities for data given fixed models (e.g. prob. of events, marginals, conditionals, etc).
- Statistics: inferring a model given fixed data observations (e.g. clustering, classification, regression).

$\theta$

Main assumption:

Independent and identically distributed random variables  
i.i.d

# Maximum Likelihood Estimation

For Bernoulli (i.e. flip a coin):

Objective function:  $P(x^{i}|\theta) = \theta^{x^{i}}(1-\theta)^{1-x^{i}}$   $x^{i} \in \{0,1\}$  or  $\{head, tail\}$

$$L(\theta|X) = L(\theta|X = x^{1}, X = x^{2}, X = x^{3}, \dots, X = x^{n})$$

i.i.d assumption

$$L(\theta|X) = \prod_{i=1}^n P(x^{i}|\theta)$$

$$L(\theta|X) = \prod_{i=1}^n P(x^{i}|\theta) = \prod_{i=1}^n \theta^{x^{i}}(1-\theta)^{1-x^{i}}$$

$$\begin{aligned} L(\theta|X) &= \theta^{x^{1}}(1-\theta)^{1-x^{1}} \times \theta^{x^{2}}(1-\theta)^{1-x^{2}} \dots \times \theta^{x^{n}}(1-\theta)^{1-x^{n}} = \\ &= \theta^{\sum x^{i}}(1-\theta)^{\sum (1-x^{i})} \end{aligned}$$

$$\log(a^b) = b \log a$$

We don't like multiplication, let's convert it into summation

What's the trick?

Take the log

$$L(\theta|X) = \theta^{\sum x^{i}} (1 - \theta)^{\sum (1 - x^{i})}$$

$$\log L(\theta|X) = l(\theta|X) = \log(\theta) \sum_{i=1}^n x^{i} + \log(1 - \theta) \sum_{i=1}^n (1 - x^{i})$$

How to optimize  $\theta$ ?

$$\frac{\partial l(\theta|X)}{\partial \theta} = 0 \quad \frac{\sum_{i=1}^n x^{i}}{\theta} - \frac{\sum_{i=1}^n (1 - x^{i})}{1 - \theta} = 0$$

$$\theta = \frac{1}{n} \sum_{i=1}^n x^{i}$$